CMSC335

Web Application Development with JavaScript

HTML I

Department of Computer Science

University of MD, College Park

Slides material developed by Ilchul Yoon, Nelson Padua-Perez

For Class Projects

- When submitting Node.js projects, remove the **node_modules** folder
- After submitting your work to the submit server, download your submission and test it works
- Keep versions of your code as you develop it. This will help you in case you submit the wrong code

IDE

- **Visual Studio Code (VS Code)** - https://code.visualstudio.com/
  - Recommended extensions: Prettier, Live/Five Server, W3C Web Validator by Celian Riboulet, ESLint (Microsoft), Live Preview (Microsoft)
    - » Live/Five Server - Right-click on HTML file and select “Open with Five Server”
    - » Live Preview - Right-click on HTML file and select “Show Preview”
      - It might not work for JavaScript
- **Let’s take a quick look at VS Code**
  - You want to open a folder where your files reside
  - The icon at the top controls the Explorer
  - Select the “Extensions” icon to search/manage extensions
    - » Let’s look for the Prettier extension
  - You can create an HTML boilerplate file by
    - » Expand the folder (select >) you have opened in VS Code
    - » Select the “New File…” option (square with a + symbol)
    - » Name the file with a .html extension
    - » Type ! at the beginning of the file and press enter

Creating Web Pages

- HTML (Hyper-Text Markup Language) created by **Sir Tim Berners-Lee**
  - https://en.wikipedia.org/wiki/Tim_Berners-Lee
- HTML Standard
  - Developed by the World Wide Web Consortium (W3C)
  - http://www.w3.org
- HTML - written in a text file
- **HTML goal is to describe structure only**. The presentation should be left to CSS (cascading style sheets)
- HTML/CSS/JavaScript/Bootstrap tutorial site
  - http://www.w3schools.com

HTML Basics

- HTML documents rely on tags
  - A tag is defined using < > and a word in between (start tag) or </> and a word in between (end tag)
  - **<title>**This is the document title**</title>**
  - Tag - surrounds the content and applies meaning to that content
  - Most HTML elements have two tags (start tag and end tag)
  - Some tags are self-closing (e.g., <br>)
  - HTML comments specified using <!-- and -->
    - » **Example:** <!-- Multiline comment-here -->
- HTML document structure
  - DOCTYPE specification
  - <html></html> tags defines the document
  - Two sections: head and body (contained within the <html></html> tags)
  - **<html>** tag may include **lang** attribute to define the language used in a document or an element <html lang="en">. For example, you can use "fr" instead of "en" for french

HTML Basics

- **head** - provides the browser's information and resources to render the page. It does not generate displayed content
  - Identified by **<head>** tag
- **body** - page contents
  - Identified by **<body>** tag
- **<meta**> - document metadata tag (e.g., charset, description, keyword)
- **<h1></h1>** - for main header specification
- ** ** - to define a paragraph
- **UTF-8** is the default HTML5 character encoding
- **Example:** HTMLTemplate1.html
- **width=device-width** - the page's width will follow the screen's width of the device
- **initial-scale=1.0** - initial zoom level when the page is first loaded

HTML Basics

- Let's validate our document both using:
  - https://validator.w3.org/#validate_by_uri
  - Using **VS Code W3C Web Validator** by Celian Riboulet
    - » Select W3C validation in status bar
    - » Expand the "PROBLEMS" area (next to OUTPUT)
- Let's introduce an error and validate
  - Although <!doctype html> validates with the external validator, it does not validate with the VS Code extension (doctype needs to be in uppercase in VS Code)

Attributes

- An attribute extends or modifies a tag
- **Attributes**
  - **Only appear in the start tag**
  - Several can appear in one tag, each separated by spaces
  - Order is immaterial
  - Some take values that are specified after an =
- **General format**
  - <elem attr="attributeValue">Displayed Text</elem>
  - **Example:**

    <img **src**="bear.gif" **alt**="bear image">
- **Boolean attributes**
  - Do not require a value
  - Represent **true** is present and **false** otherwise
  - **Example:** <input type="checkbox" name="room" **checked**>

Elements in <head>

- **<title>**
  - **<title></title>**
  - Required and search engines rely on it
  - Provide a meaningful name, as it is bookmarked
- **<meta>**
  - Document metadata (e.g., keywords, description, encoding, refresh cycle, etc.)
- **Viewport:** user's visible area of a web page. Generally, the same as the browser's window
  - The following should be part of every page:
    - » **<meta name="viewport" content="width=device-width, initial-scale=1.0">**
  - **width=device-width -** the page's width will follow the screen's width of the device
  - **initial-scale=1.0 -** initial zoom level when the page is first loaded

Elements in <head>

- **<link>** - Specifies the relationship between the current document and an external resource
    - » Common uses: to add style sheets and favicons
    - » Favicon - browser tab icon. Small icon representing your page
    - » Favicon generator - https://realfavicongenerator.net/
- **<style>** - To define CSS style in the document
- **Example:** HTMLTemplate2.html
- **File types** (mime types)
  - **MIME** (Multipurpose Internet Mail Extensions) - standard to indicate the nature of a file, document, or group of bytes
  - In the previous example **image/png** is a MIME type
  - https://mimetype.io/all-types
- **Example:** RefreshingPageWithMetaTag.html

Nested Tags and Spaces

- Nested tags are possible (but keep the correct order)
  - **Incorrect -** Message
  - **Correct** - Message
- Browser processing of spaces - multiple spaces are converted to one space

  John Mary Peter

  will be converted to:

  John Mary Peter
- Line returns are ignored
- The same rule regarding spaces applies to text appearing in between tags
  - <h1>There is a bird</h1> - only one space in-between **there** and **is**
- Let's see an example
  - We can use <pre> to preserve the spaces

Frequently Used Tags

- **Heading:** <h1> thru <h6>
- **Paragraph: ** text here ** **
- **Line Breaks**:
- **Emphasized Text **
  - **** text here **** *Text is usually rendered in italics*
- **Important Text **
  - **** text here ** Text usually rendered in bold**
- **Super/Subscript**
  - **_{** text here **}**
  - **^{** text here **}**
- **Preformatted text <pre>**
  - Spacing of text is respected (NOT verbatim)
  - **<pre>** The space will appear**</pre>**
  - You can have tags inside them
- **
 -** Single line break
- **Example**: FrequentlyUsedTags.html

Character Entity References

- **Special Characters (Character Entity References)**
  - Non-break space: ** ** (allows us to add a single space in the document)
  - Copyright: **©**
  - Registered Trademark: **®**
  - Ampersand: **&**
  - Less than (<): **<**
  - Greater than (>): **>**
    - » The previous two allow us to display a tag on a page (e.g., <p>)
- **Unicode**
  - You can display **Unicode** characters as well
  - E.g., **♛** will display ♛
  - E.g., **⚀** will display ⚀
- Unicode Site: https://home.unicode.org/
- List: https://tools.w3cub.com/html-entities
- **Example:** CharacterReferences.html

HTML Lists

- **Unordered lists**
  - <ul></ul> identify the list
  - <li></li> each list item
- **Ordered lists**
  - <ol></ol> identify the list
  - <li></li> each list item
  - **start** attribute used to define the starting point
  - **reversed** attribute used to reverse order
  - **type** attribute to define item type (e.g., A, a, I, I, 1)
- **Definition lists**
  - <dt></dt> - term
  - <dd></dd> - description
- **Lists can be nested**
  - Notice where the text inside of the main list appears
  - Notice the different symbols for the unordered list that is nested
- **Example:** Lists.html

HTML Image Inclusion

- **<img>** tag used for image inclusion
- Some <img> attributes
  - **src**
  - **width (optional)** and in pixels
  - **height (optional)** and in pixels
  - **title (optional)**
  - **alt**
- **<img>** are rendered next to each other on the same line (inline element)
- To reduce the image size without losing proportions, specify either the **width** or **height**, and the browser will compute the other dimension
- **Example:** Images.html
- **<hr>** - thematic change, rendered as a horizontal rule
- Image can be online (not necessarily local)

HTML Hyperlink (anchor <a> tag)

- Hyperlink: connection between web resources
- Links are created using the anchor **<a>** tag
  - <a href="http://www.cnn.com">CNN Page</a>
  - You need to specify http:// (protocol)
  - You can add a **title** using title="News"
- You can also link a local page with another
  - <a href="page.html">page.html</a>
- You can create links (bookmarks) to elements in the same document using the **id** attribute
- You can turn an image into a link by replacing the text in <a></a> with an <img>
- **A reference to the same page <a href="#"></a>**

HTML Hyperlink (anchor <a> tag)

- **Example:** Hyperlinks.html
  - Adding the **download** attribute downloads instead of rendering the file
  - You need to use a web server to see the effect (you will not see it by just opening the file with the browser)
- Example that shows the **download** attribute (including renaming downloaded file)
  - https://www.cs.umd.edu/~nelson/classes/resources/HTML5CSS3/downloadLink/
- **<a>** can include a **target** attribute that indicates the destination where the document will be opened: **_blank** (new window), **_self** (default), **_parent** (parent frame), **_top** (current window)